SkylineOS

Student Organization System: one platform for events, members and money

Hackathon prep pack: literature review, market scan, feature plan, review Q&A and 24-hour team split

Contents

1. Problem in one page (scenes mapped to modules)
2. Literature review (paper, authors, summary, gap)
3. Current market alternatives
4. Core features (must-have) and wow features (to win)
5. Suggested tech stack and data model
6. Questions the reviewers may ask, with strong answers
7. Team split for 4 people, 24-hour timeline, demo script and cut-list

1. Problem in one page

The Skyline Student Association runs on a spreadsheet (members), cash at the door (tickets), a notebook (budget), WhatsApp (announcements) and chased receipts (reimbursements). The result: members lose track of events, money goes missing and nobody knows who owes what. The brief asks for **one unified platform**, so the winning idea is not six separate tools but one shared member identity and one shared ledger that every module writes to.

Scene in the brief	Real pain today	Module we build	Writes to ledger?
A new student joins	Spreadsheet sign-ups; no renewal reminders; cannot verify membership at events	Membership: plans, dues payment, digital member card with QR, renewal engine	Yes (dues)
Selling tickets for the spring gala	Cash at the door; no seat limit; no printed list tolerance	Events: member vs non-member pricing, capacity and waitlist, QR ticket, door scanner, post-event report	Yes (ticket revenue)
Announcing the next meeting	Same message pasted in several WhatsApp groups	Announcements: post once, publish to web, email and share links; searchable archive	No
Ordering hoodies	Sizes noted on paper; payment chasing; unknown stock	Merch shop: size variants, stock counts, member discount, online payment, order status	Yes (merch revenue)
Planning a fundraiser	Tasks live in someone's head or a group chat	Projects: task board, owners, due dates, progress meter, on-the-day sales log	Yes (fundraiser income)
Treasurer at semester end	Notebook and a pile of receipts	Finance: ledger, expense claims with receipts and approval, dashboard, one-click report	Source of truth

2. Literature review

Academic work in this space is mostly **single-purpose**: QR attendance, event registration or club finance, studied in isolation. The table lists the papers and project reports found during this research, what each did, and the gap that SkylineOS fills.

> *Check before citing: details below come from abstracts and web pages found in search. Authors or years that were not visible are marked "verify". Open each link once and confirm the details before you put them on a slide.*

#	Paper / report	Authors, venue	Summary	Gap we address
1	Location-aware Event Attendance System using QR	IJACSA (SAI) Vol. 9 No. 9, Paper 59. Authors: verify on the PDF	Android app for university events. Organizer creates an encrypted QR for an event; students log in and scan; GPS location is checked to confirm presence.	Attendance only. No ticket pricing, memberships, payments, stock or finance. Needs connectivity and a student login.
2	Event Attendance System for SIRIM Berhad	S. Shukri, N. M. Nawi, Universiti Tun Hussein Onn Malaysia (AITCS journal)	Web system replacing manual attendance with QR scanning. Modules: authentication, event management, real-time attendance, registration, analytics, payment, reminders.	Built for one corporate body. No member tiers or discounts, no merch inventory, no volunteer tasks, no club ledger or reimbursement flow.
3	Automated QR Code Attendance Tracking System for Efficient Classroom Management	G. Ambaye, Anderson, Y. Yihun, ASEE (Paper ID 53751)	Per-student QR codes scanned by camera; evaluated with students, 70% completed check-in in under ten seconds. Operator runs it from a personal device.	Classroom scope. Single operator device, no offline mode, no ticket validity or capacity logic, no payments.
4	Smart Attendance System for University Students Based on QR Code	A. Sneineh, W. A. Salah, J. of Electrical and Electronics Eng., Vol. 18 No. 1, 2025	ID card QR read by a camera connected to a PIC microcontroller and PC; matches student to a section database and logs attendance. Reported time savings and fewer errors.	Hardware dependent and fixed-location. Nothing about events, money or members. Our scanner is just a phone browser.
5	College Event Management System	M. I. Iskak, H. M. Ekhsan, UiTM, REMACS 4.0 abstract book, 2022	Web system using QR for faster event registration; built with the Waterfall model and tested with functionality and user acceptance tests. Future work: a mobile app.	No pricing tiers, payments, capacity control or finance. The mobile app they list as future work is a PWA in our build.
6	Enhanced QR-code Based Attendance System (user-centric approach)	Nigerian university authors, J. of Computer Science and Information (UNIMAS). Authors: verify	Improves QR attendance with identity authentication and privacy focus; reports 7,900 unique scans over seven weeks.	Strong on scale and privacy but only attendance. We reuse the privacy idea: signed tokens, no personal data in the QR.
7	BEST Skopje Hub: Integrated Membership, Activities and Event Management System	T. Bogoeski, FINKI Skopje, Databases course 2025/26 project wiki	Database design for one student organisation: membership history (observer to alumni), time-limited board roles, RSVPs, mandatory attendance, applications and mentorships.	Excellent membership-lifecycle model, but no ticket sales, merchandise, fundraisers or financial ledger. We borrow its idea of time-limited roles for yearly elections.
8	Brown University Student Organization Information and Financial Management System	Brown CS190 project proposal from the Student Activities Office and Finance Board (2005 era)	Describes paper budget requests, the same data typed into several systems and paper membership lists; asks for a central system with live budgets and exportable rosters.	Shows the problem is old and finance-centred. We deliver the proposed central roster and real-time budget, plus events and shop on top.
9	Treasurer handbooks (Purdue Extension 4-H Treasurer's Handbook; Univ. of Kentucky CLD2-11; UW Extension Club Treasurer guide)	Extension services, not peer-reviewed; used as practice evidence	Recommend approved budgets, receipts for every reimbursement, a reimbursement form and a written monthly report to members.	All manual forms and ledgers. We turn these rules into software: approval before spend, receipt required, auto monthly report.

What the literature tells us

- **QR check-in is a proven, accepted pattern** (rows 1 to 6), so we do not need to justify it, only do it better: offline, signed and fast.
- **Nobody combines attendance, money and membership.** Papers either cover attendance, or events, or finance. Our claim: one identity and one ledger across all six scenes.
- **Finance practice is rule-based** (row 9), which makes it ideal to encode: approvals, receipts, monthly reports.
- **Membership lifecycle matters** (row 7): statuses, renewals and yearly role handover are first-class data, not an afterthought.

3. Current market alternatives

Tools exist for each scene, but a club ends up stitching three or four of them together. Pricing is indicative and from public listings; check current plans before quoting.

Tool / stack	Type	What it does well	Where it falls short for Skyline
Campus platforms (CampusGroups, InvoNet, Argo Pulse)	University-licensed org management	Org directories, rosters, event check-in apps, email messaging, funding requests and budgets	Tied to an institution and admin-access approval; clubs cannot run their own tiered pricing, merch or volunteer tasks; finance is request-based, not a live ledger
Join It	Club membership and dues	Mobile-first; membership, events and dues; automated renewals and payment reminders	Strong on members, lighter on merch stock, fundraiser tasks and a treasurer ledger
WildApricot	Membership management SaaS	Member database, event registration, email, website, payments; entry plan listed from about US$124.95 for 500 contacts (G2)	Priced in USD for associations; reviewers call event management inflexible; no volunteer task board or expense approvals
ClubRunner, ClubExpress, ClubDesk	Association / club management	Dues, invoices, committees, calendars, websites	Built for service clubs and associations; heavy for a student club; no offline QR door flow
Eventbrite and similar ticketing	Ticketing	Polished ticket sales, attendee lists, email	No membership link, so member pricing is manual; fees per ticket; no ledger
Volo Cash, Xpnse AI	Volunteer treasurer expense tools	Expense submissions, approvals, budget tracking for volunteer-run orgs and student clubs	Finance only; cannot see ticket, dues or merch income automatically
Google Forms + Sheets + UPI/Razorpay links (typical Indian club stack)*	DIY	Free and familiar	Manual reconciliation, no capacity locking, no QR validation, no audit trail. This is the real competitor on a campus.
WhatsApp groups + notebook (status quo)	Manual	Zero setup	This is the problem in the brief

** Not verified in this research round; included from general knowledge of how campus clubs operate.*

Our positioning

Dimension	Typical alternative	SkylineOS
Scope	One or two scenes per tool	All six scenes on one member identity
Money	Separate payment tool; manual reconciliation	Every payment and expense posts to one tamper-evident ledger
Door check-in	Printed list or online-only scanner	Offline-capable PWA scanner with signed QR; any phone works
Pricing logic	Manual discount codes	Automatic member-tier pricing across tickets and merch
Cost and fit	USD subscription, association-style	Free for a student club; UPI-ready; yearly leadership handover built in

4. Features

4.1 Core features (must be in the demo)

Module	Feature	Notes for the build	Pri
Auth and roles	Sign up and log in; roles: Member, Volunteer, Door staff, Treasurer, Admin	Role-based access on every API route; Treasurer sees finance, Door staff sees only scanner	P0
Membership	Plans (for example Standard, Premium), dues payment, status Active / Expiring / Lapsed, expiry date, benefits shown on the member card	Discount percent stored on the plan and read by tickets and shop	P0
Membership	Digital member card with QR; admin member list with search, filters and CSV export; renewal reminders at 30, 7 and 1 days	Reminders via scheduled job and email	P0

Module	Feature	Notes for the build	Pri
Events	Create event with date, venue, ticket types, member and non-member price, seat limit	Seats counted inside a database transaction so overselling is impossible	P0
Events	Online checkout; ticket with QR; "seats left" counter; waitlist when sold out	Payment gateway in test mode	P0
Door check-in	Phone scanner: green for valid, red for invalid, amber for already used; scan member card or ticket	Result in under a second; duplicate scan blocked	P0
Event report	Registered vs attended, no-show rate, revenue, ticket mix, member vs non-member split	Auto-built from tickets and check-ins	P0
Announcements	Post once; appears on public site, member feed and email; stored with timestamp and author; searchable archive	Pinned posts for deadlines	P0
Merch shop	Products with size variants, stock count, member price, cart, payment, order status (Paid, Ready, Collected)	Stock decrement inside a transaction; low-stock flag	P0
Fundraiser	Task board (To do, Doing, Done) with owner and due date; progress bar; income and cost for the fundraiser	One project page showing "on track / at risk"	P0
Finance	Ledger of all income and expenses by category and event; expense claim with receipt photo; approve or reject; reimbursed status	Dashboard: in, out, balance, who is still owed	P0
Finance	Semester report: dues, tickets, merch, fundraisers, expenses, closing balance; exports as PDF and CSV	The treasurer scene is the closing demo moment	P0

4.2 Wow features (what makes judges remember us)

Pick the top five or six; effort is a rough estimate for one person. Anything marked **Do** is recommended for the 24-hour window.

Wow feature	What it does	Why it wins	Effort	Call
Offline-first signed QR scanner (PWA)	Tickets and member cards carry a signed token (no personal data inside). The scanner caches the event list, validates on the phone with no network, and syncs check-ins when back online. Duplicate scans are caught across devices on sync.	Directly answers "check a ticket in seconds without a printed list", even with bad venue Wi-Fi. Very demo-able: switch airplane mode on stage.	4 to 5 h	Do
Rotating QR on member card	Card QR refreshes every 30 seconds, so screenshots of a card cannot be shared.	Shows security thinking; ties to "check that they are in fact a member".	1 h	Do
Receipt scanner for expenses	Volunteer photographs a receipt; a vision model extracts vendor, date and amount, pre-fills the claim and flags mismatches with the amount entered.	Kills the "chasing receipts" pain in the brief. Visible AI that has a clear job.	3 h	Do
Tamper-evident ledger	Ledger entries are append-only; each stores a hash of the previous one; an "Audit" page verifies the chain and shows edit history.	Answers "money goes missing" with proof, not trust.	2 h	Do
Money-flow dashboard and ask-the-books	Sankey of income sources to spend categories; a natural-language box ("How much did the gala make after expenses?") answered from read-only queries.	Treasurer scene becomes a showpiece; "at a glance" literally met.	3 to 4 h	Do
Smart pricing engine	One rules table drives member discount, early-bird window, and group rate for tickets and merch.	Shows the unified design: change a plan once, prices change everywhere.	2 h	Do
Waitlist with auto-promotion	When a ticket is cancelled or a hold expires, the next person on the waitlist gets an email and a 15-minute pay window.	Handles the real gala scenario; shows concurrency thinking.	2 h	Stretch
Cart hold for hoodies	Stock is reserved for 10 minutes during checkout and released if unpaid.	Prevents "paid but out of stock".	1.5 h	Stretch
One post, many channels	Announcement publishes to site, email and gives a pre-formatted WhatsApp share link plus an optional Telegram bot; shows read count.	Replaces the repeat-paste workflow while respecting that WhatsApp is where students are.	2 h	Stretch

Wow feature	What it does	Why it wins	Effort	Call
Churn-risk and engagement score	Score per member from attendance and purchases; renewal nudge prioritises at-risk members.	Turns data into action for the leadership team.	2 h	Stretch
Leadership handover wizard	Elected officers take over roles on a date; old officers lose finance access; the ledger and history stay.	Brief says leadership is elected every year; few teams will cover it.	1.5 h	Stretch
Live semester simulation	Seed script plays a whole semester (sign-ups, gala, hoodies, bake sale, reimbursements) so every dashboard is full in the demo.	Empty dashboards lose hackathons.	1.5 h	Do

5. Suggested tech stack and data model

Layer	Choice	Reason
Frontend	Next.js (React) + Tailwind; installable PWA	One codebase for site, admin and scanner; PWA gives offline and camera access
Backend and DB	Next.js API routes or a small Node service; PostgreSQL (Supabase or similar) with row-level security	Transactions for seats and stock; managed auth and storage for receipts
Payments	Razorpay or Stripe in test mode; UPI collect or payment links; webhooks mark orders paid	Fits Indian campus use; webhook-driven so the ledger is correct even if the browser closes
QR and tokens	Signed tokens (HMAC or Ed25519), QR via a JS library, scanner via the browser camera API	Verify without network; no personal data inside the code
Email and jobs	Transactional email API; scheduled jobs for renewals and waitlist expiry	Reminders and announcements
AI	Vision-capable model for receipts; text-to-SQL limited to read-only views for the books	Small, safe, visible features
Deploy	Vercel or similar plus managed Postgres; seeded demo database	Live URL for judges

Core entities

Entity	Key fields	Links
users	id, name, email, role, college id (optional)	has memberships, tickets, orders, claims
membership_plans / memberships	plan: name, price, discount %, duration; membership: user, start, end, status	plan drives pricing rules
events / ticket_types / tickets	event: date, venue, capacity; type: member price, public price, quota; ticket: owner, status, qr token, checked_in_at	ticket links to order and ledger
products / variants / orders / order_items	variant: size, stock, price; order: status, total	order links to ledger
announcements	title, body, author, published_at, channels	read receipts per member
projects / tasks	project: goal amount, due date; task: owner, status, due	project links to ledger entries
ledger_entries	id, type (in/out), category, amount, source (ticket, dues, merch, fundraiser, expense), ref id, prev_hash, hash, created_at	single source of truth for finance
expense_claims	claimant, amount, receipt url, status (Submitted, Approved, Rejected, Reimbursed), approver	approval posts an "out" entry
audit_log	actor, action, entity, before/after, time	shown on the Audit page

6. Questions reviewers may ask

Rehearse these. Answers are written for the design in this document; adjust if your final build differs.

Problem and product

Question	Strong answer
Why build one platform instead of using existing tools together?	Because the pain is the gaps between tools. A hoodie bought by a member should change stock, the member's discount, the ledger and the treasurer's report in one step. With separate tools someone reconciles by hand, which is today's problem.
Who is the primary user?	Three users: the member (join, buy, check in), the volunteer (tasks, claims) and the treasurer (the books). The treasurer's time-saving is the strongest value claim, so the demo ends there.
What makes this different from WildApricot or Join It?	They are subscription tools centred on members. We unify members, door check-in, merch, fundraiser tasks and a tamper-evident ledger, free for a student club, with UPI support and yearly leadership handover.
How do you measure success?	Door check-in time per person, percentage of renewals on time, time to produce the semester report (hours to under a minute), and number of unreconciled payments (target zero).
What did you deliberately leave out?	Full accounting (GST, tax filing), multi-club tenancy, native apps. We kept scope to the six scenes and made them deep.

Technical design

Question	Strong answer
How do you stop overselling seats or hoodies when many people buy at once?	Seat and stock decrements happen inside a database transaction with a row lock (or an atomic conditional update). A hold with expiry covers the checkout window; a scheduled job releases expired holds.
How does the door scanner work with bad Wi-Fi?	It is a PWA that caches the event's signed ticket list and public verification key. It validates the signature and status locally, stores check-ins on the device and syncs later. Conflicts (same ticket scanned on two phones) are resolved by earliest timestamp and flagged.
What is inside the QR code?	Only a ticket or member id, an event id, an expiry and a signature. No name or email, so a leaked QR image exposes nothing, and a forged one fails verification.
How do you prevent a member card or ticket screenshot being shared?	Member card QR rotates every 30 seconds; tickets are single-use and the second scan shows "already used" with the time and gate.
How is the ledger tamper-evident?	Entries are append-only. Each entry stores a hash of its content plus the previous hash, like a simple chain. The Audit page recomputes the chain; any edited row breaks it. Corrections are new reversing entries.
Why a webhook for payments?	The browser can close after paying. The gateway's signed webhook is the source of truth: it marks the order paid, issues the ticket and posts the ledger entry, and is idempotent so retries do not double-post.
How does the AI receipt reader avoid wrong amounts?	It only pre-fills; the volunteer confirms. If the extracted amount differs from the typed amount, the claim is flagged for the treasurer. The "ask the books" box runs only on read-only views and shows the SQL it used.
What is your scaling story?	A club has hundreds to a few thousand members. Postgres on a managed plan handles this easily; the hot path is ticket sales on gala day, which is one indexed transaction per purchase. Static pages and the scanner cache reduce load.

Security, privacy and trust

Question	Strong answer
How do you handle personal data and student privacy?	Collect only name, email and optional college id. Role-based access and row-level rules; Door staff see only a green/red result and a first name. Receipts live in private storage. Members can request export or deletion.

Question	Strong answer
What stops a volunteer approving their own claim?	Separation of duties: the claimant cannot approve their own claim, and claims above a threshold need two approvers (treasurer plus president). Every action is in the audit log.
What if the treasurer's account is compromised?	Optional second factor for finance roles, short sessions, and the hash-chained ledger means silent edits are detectable. Reimbursement payout is a separate confirmed step.
How do you protect payments?	We never touch card data; the gateway hosts the payment form. We verify webhook signatures and store only gateway reference ids.

Business, impact and roadmap

Question	Strong answer
How would this make money or sustain itself?	Free for a single club; paid tier for multi-club federations and campus offices (analytics, approvals, payouts). Alternatively a small convenience fee passed to non-members on tickets.
How would other clubs adopt it?	Multi-tenant by club id (already in the schema), one-click import of the old spreadsheet, and the leadership handover wizard so the next committee inherits everything.
What would you build next?	Native wallet passes, multi-club tenancy, bank-feed reconciliation, UPI auto-payout of reimbursements, and sponsor management.
What are the biggest risks?	Payment edge cases and offline conflicts. We mitigate with webhook idempotency, test-mode runs of failure cases and the earliest-scan-wins rule.

Team and process

Question	Strong answer
How did you split the work in 24 hours?	Schema and API contract agreed in hour one, then four vertical owners: backend and data, member-facing UI, door and communications, finance and AI plus pitch. Integration checkpoints at hours 8, 14 and 20; feature freeze at hour 20.
What did you cut and why?	See the cut-list in section 7: we protected the end-to-end path (join, pay, scan, ledger, report) and dropped polish features first.
How did you test it?	Seeded semester dataset, a script that buys the last seat from two sessions at once, webhook replay tests and a rehearsed offline scan on a real phone.

7. Team split for 4 people, 24 hours

Rule of thumb: **one vertical slice per person**, so nobody waits for anyone else. Agree the database schema and API contract in the first hour. Names are placeholders; swap roles to fit strengths.

Person	Role	Owns	Key deliverables
A	Backend and data lead	Schema, auth and roles, pricing engine, payments and webhooks, transactions for seats and stock, ledger posting	Running API with seed data; checkout that issues tickets and orders and posts to the ledger; role guards
B	Frontend and member experience	Landing site, sign-up, membership purchase, member card, events and ticket purchase, merch shop and cart, announcements feed, shared design system	Complete member journey on mobile; clean, consistent UI; empty and error states
C	Door, comms and projects	QR signing and verification, offline PWA scanner, attendance analytics, announcement publishing and email, renewal and waitlist jobs, fundraiser task board	Scanner that works in airplane mode; post-event report; announcement archive; bake-sale board with progress

Person	Role	Owns	Key deliverables
D	Finance, AI and pitch	Expense claims and approvals, receipt scanner, treasurer dashboard, semester report PDF, hash-chain audit page, ask-the-books, seed script, deck and demo	Treasurer dashboard and report; wow features 3 to 5; demo script, slides, backup video

Hour-by-hour plan

Hours	Everyone	A (Backend)	B (Frontend)	C (Door and comms)	D (Finance and pitch)
0 to 1	Kickoff: lock scope (P0 list), schema, API contract, repo, branches, deploy skeleton	Create DB and auth	Design tokens, layout, routing	Pick QR and PWA libraries, token format	Seed data outline, deck skeleton
1 to 6	Short stand-up at hour 3	Tables, roles, plans, events, ticket types, pricing engine	Landing, sign-up, plans page, member card (static)	Token signing, scanner UI with camera, announcement CRUD	Ledger table and posting API, expense claim form and storage
6 to 10	Checkpoint at hour 8: demo join and buy	Payments, webhooks, seat and stock transactions	Event page, checkout, merch shop, cart	Scan result states, check-in table, email sending	Approval workflow, dashboard v1 (totals, balance)
10 to 14	Checkpoint at hour 14: full path works end to end	Renewal job, member admin list and CSV, bug fixes	Admin screens: events, orders, members	Offline cache and sync, event report, task board	Semester report PDF, receipt scanner v1
14 to 15	Rest block 1: A and B take a short break or nap (about 1 hour); C and D continue				
15 to 16	Rest block 2: C and D take a short break or nap (about 1 hour); A and B continue				
16 to 20	Wow features, then feature freeze at hour 20	Waitlist and holds; rotating QR token endpoint	Polish, dark mode, mobile fixes, loading states	Multi-channel announcements, fundraiser progress	Money-flow chart, audit page, ask-the-books
20 to 22	Bug bash with fresh eyes: each person tests another person's module	Concurrency test, webhook replay	Cross-browser and phone testing	Real-phone offline test	Seed full semester, deploy final
22 to 23	Rehearse demo three times, assign speaker per scene				Record backup video, finish slides
23 to 24	Buffer, submit, check live URL, charge laptops and phones				

Cut-list (drop in this order if time runs short)

1. Churn-risk score and engagement leaderboard
2. Telegram bot and read counts (keep email and the WhatsApp share link)
3. Leadership handover wizard
4. Cart hold for hoodies (keep the transactional stock decrement)
5. Waitlist auto-promotion (keep a plain waitlist)
6. Ask-the-books natural-language box (keep the dashboard)

Never cut: join and pay, ticket purchase with seat limit, QR scan with valid/invalid/used, merch with sizes and stock, announcement archive, task board, expense claim with approval, ledger dashboard and semester report.

Demo script (about 5 minutes)

Time	Scene	What to show	Line to land
0:00	Hook	Photo of the notebook, spreadsheet and WhatsApp chaos; then the SkylineOS home screen	"Members lose track, money goes missing. We fixed it with one member identity and one ledger."
0:30	New student joins	Sign up, pay dues, member card appears with rotating QR	Plan sets discounts everywhere.

Time	Scene	What to show	Line to land
1:15	Gala tickets	Member price vs public price, seats left counter, buy the last seat from two phones	Overselling is impossible.
2:00	Door	Turn on airplane mode; scan valid, used and forged QR	Seconds, no printed list, no internet.
2:30	Announcement and hoodies	One post to site and email; order a hoodie in size M as a member	Stock and ledger update instantly.
3:15	Fundraiser	Bake-sale board with owners and progress bar	On track at a glance.
3:45	Treasurer	Volunteer photographs a receipt, claim is pre-filled and approved; dashboard shows in, out, balance; audit page verifies chain; export report	"One place that shows what came in, what went out and how much is left."
4:45	Close	Roadmap and ask	Free for any club; multi-club next.

Definition of done before submission

- Live URL works on a phone; demo accounts for Member, Door staff and Treasurer are written on a card.
- Database is seeded with a full semester so every chart has data.
- Payments run in test mode and the webhook flow has been replayed once.
- Offline scan tested on a real phone; backup demo video is recorded.
- README lists features, stack, how to run, and the answers to the top ten review questions.

Sources used in the literature and market scan

- IJACSA, "Location-aware Event Attendance System using QR" (SAI conference downloads, Vol. 9 No. 9, Paper 59)
- UTHM AITCS, "Event Attendance System for SIRIM Berhad" (Shukri and Nawi)
- ASEE Paper ID 53751, "Automated QR Code Attendance Tracking System for Efficient Classroom Management" (Ambaye et al.)
- DOAJ listing, "Smart Attendance System for University Students Based on QR Code" (Sneineh and Salah, 2025)
- UiTM REMACS 4.0 abstract book, "College Event Management System" (Iskak and Ekhsan, 2022)
- UNIMAS JCSI, "Design and Implementation of an Enhanced QR-Code Based Attendance System"
- FINKI Skopje project wiki, "BEST Skopje Hub" (Bogoeski)
- Brown CS190 project idea, Student Organization Information and Financial Management System
- Purdue Extension 4-H Treasurer's Handbook; University of Kentucky CLD2-11; UW Extension club treasurer guide
- Join It blog on student organization management systems; G2 and Capterra pages for WildApricot, ClubRunner, ClubExpress, ClubDesk, Volo Cash; Product Hunt page for Xpnse AI; UWF Argo Pulse, UT Toledo InvoNet and RIT campus pages